ECLR16 - COMMUNICATION ENGINEERING LABORATORY

Sampling, Aliasing, and Signal Representations

Sampling

Sampling of Analog Signals

- $x_a(t)$ be the analog signal, the discrete time signal is obtained by taking samples every T seconds.

$$x(n) = x_a(nT), -\infty < n < \infty$$

T – time interval between successive samples – sampling period

sampling rate $F_s = 1/T$

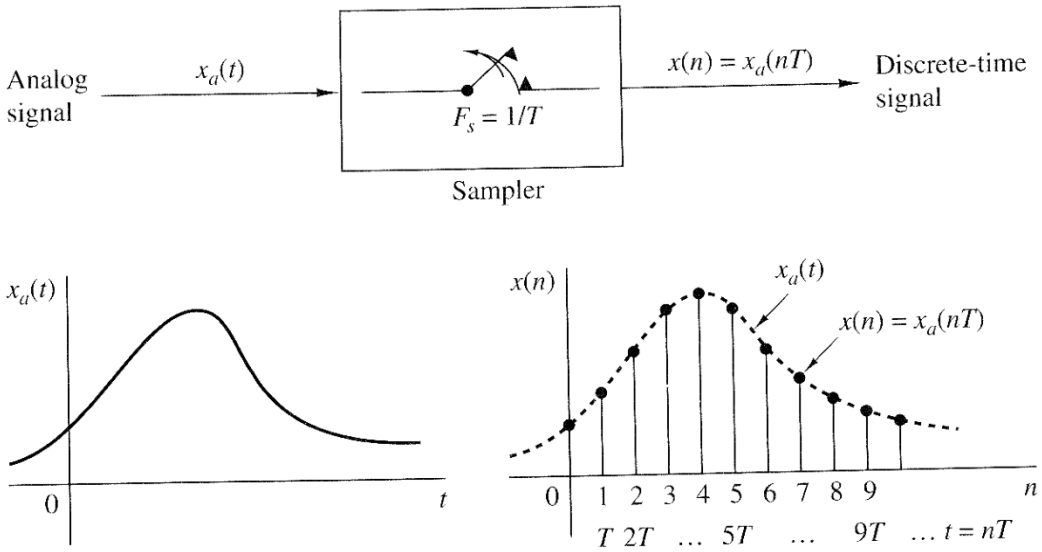


Figure 1.4.3 Periodic sampling of an analog signal.

p.21 Proakis, Manolakis, Digital signal processing: Principles, algorithms and applications – 4th edition

The Sampling Theorem

- Consider an analog signal represented by sum of sinusoids,

$$x_a(t) = \sum_{i=1}^{N} A_i cos(2\pi F_i t + \theta_i)$$

- Then to avoid problem of aliasing, the sampling rate is chosen as

$$F_i > 2F_{max}$$

where,

F_{max} is the largest frequency component in the analog signal.

$$F_{max} = \max\{F_1, F_2, \dots \dots, F_N\}$$

p.27,p.387 Proakis, Manolakis, Digital signal processing: Principles, algorithms and applications – 4th edition

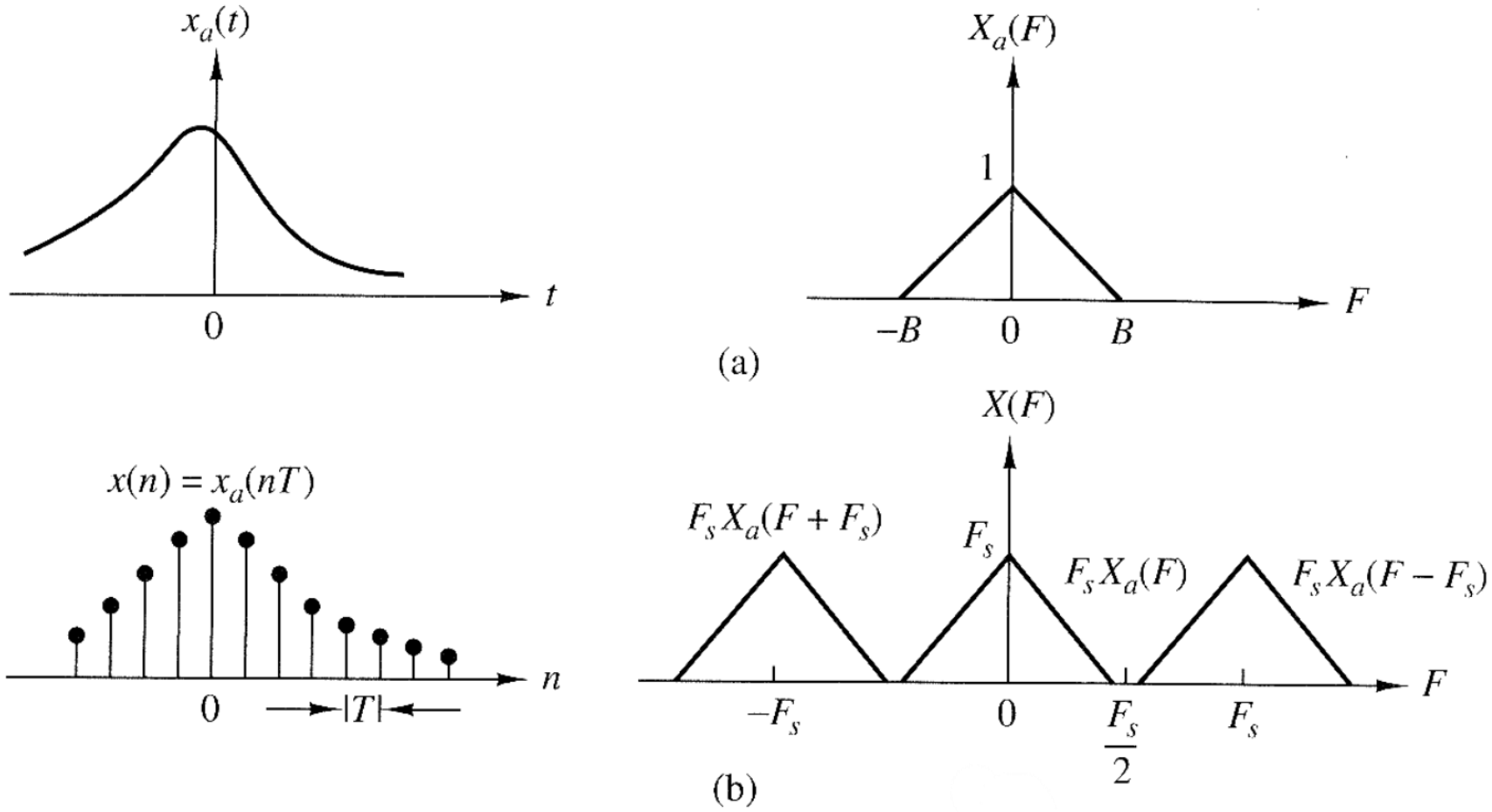
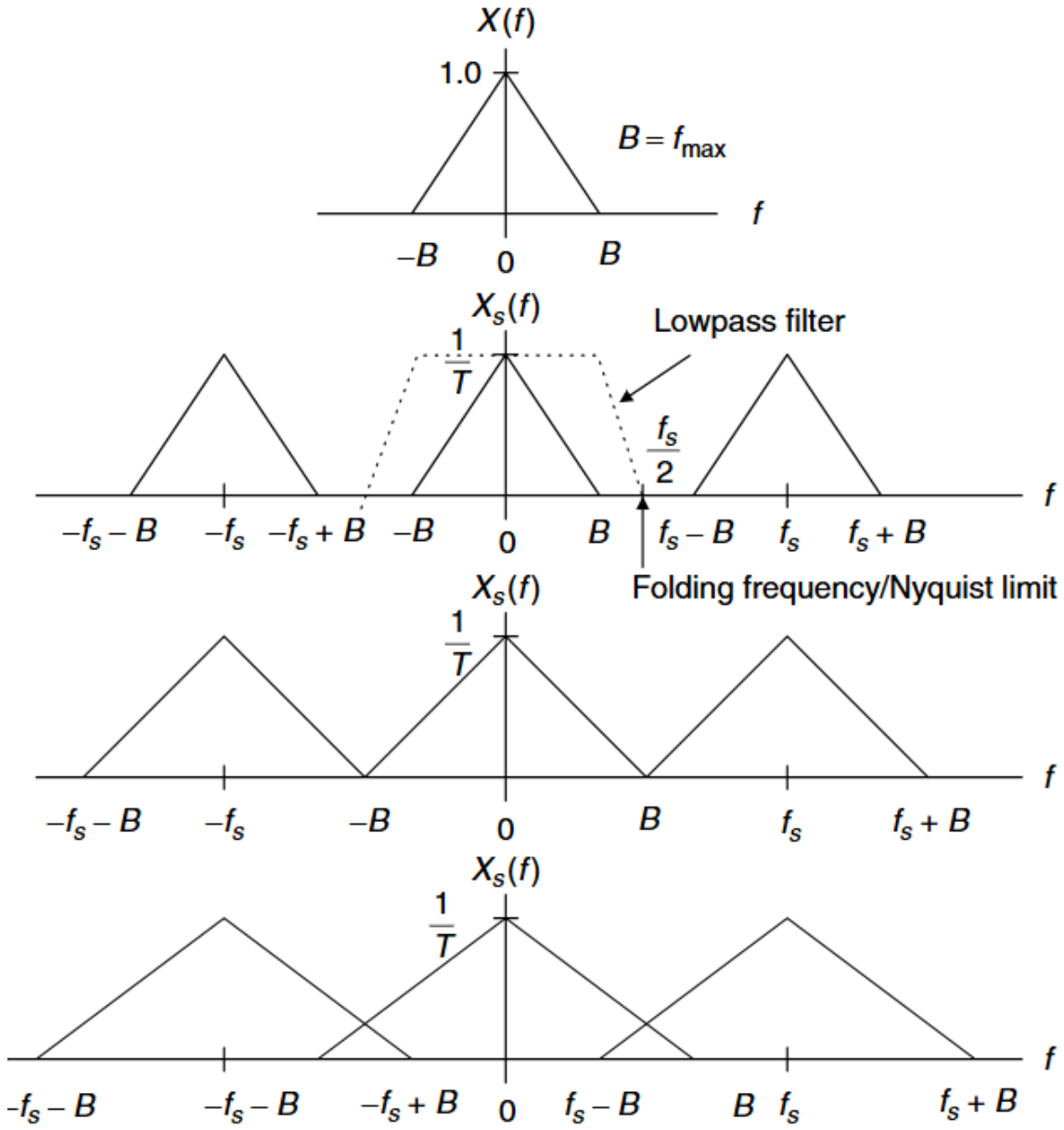


Figure 6.1.1 Sampling of an analog bandlimited signal and aliasing of spectral components.

p.388 Proakis, Manolakis, Digital signal processing: Principles, algorithms and applications – 4th edition



p.18 – Fig 2.6 Li Tan, Digital signal processing: Fundamentals and applications.

Hands-on

Aliasing

- Aliasing occurs when the signal is sampled below the Nyquist rate.
- Overlapping of spectral components – high frequency component appear as low frequency component

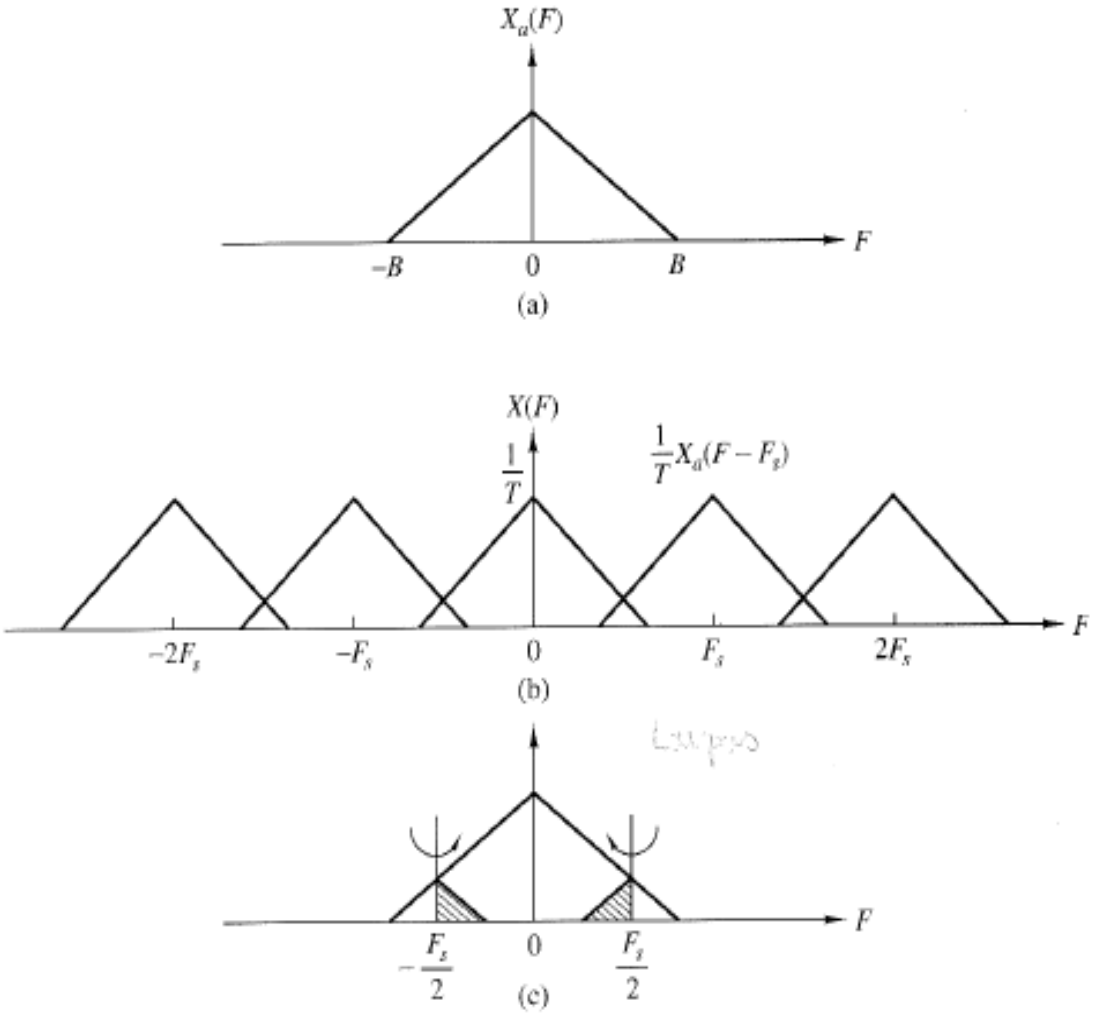


Figure 6.1.3 Illustration of aliasing around the folding frequency.

p.390 Proakis, Manolakis, Digital signal processing: Principles, algorithms and applications – 4th edition

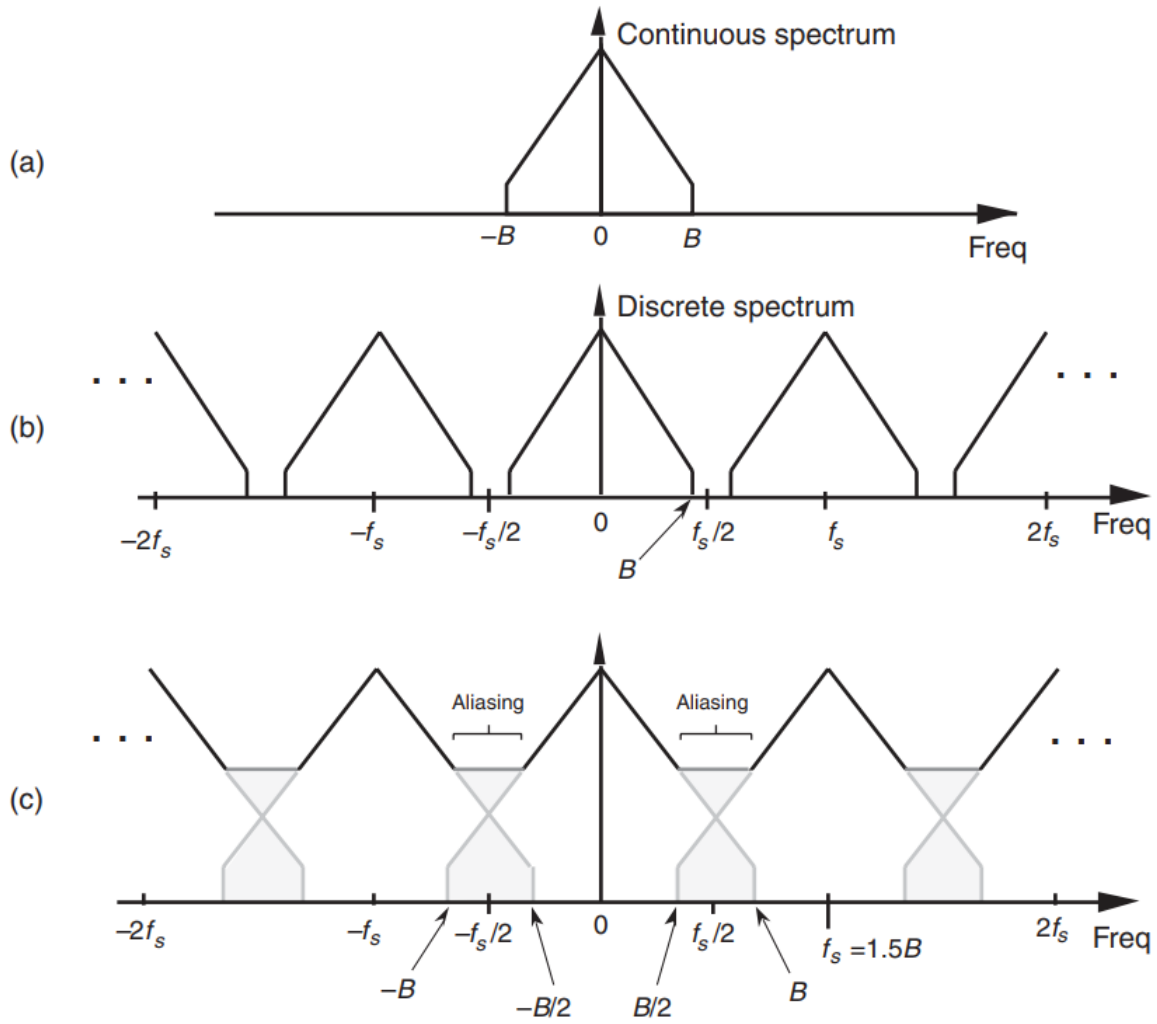


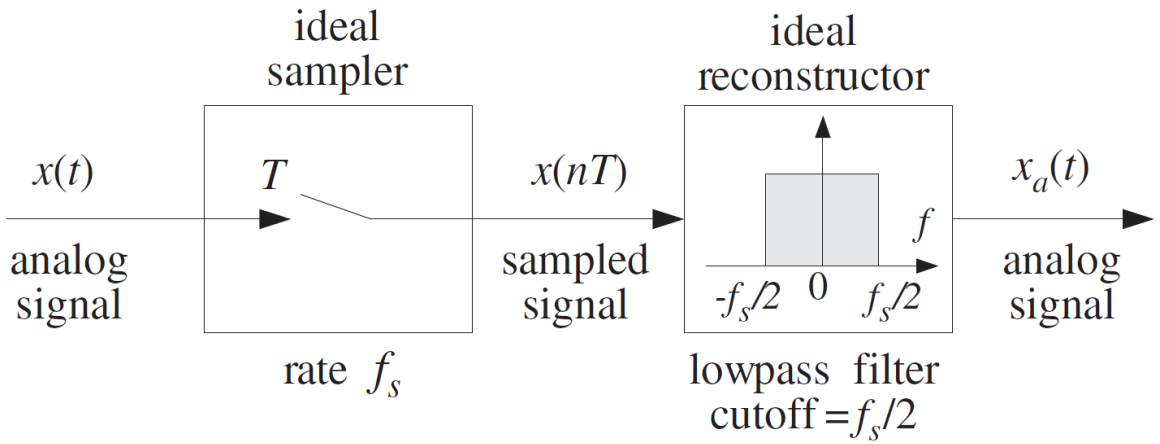
Figure 2-4 Spectral replications: (a) original continuous lowpass signal spectrum; (b) spectral replications of the sampled lowpass signal when $f_s/2 > B$; (c) frequency overlap and aliasing when the sampling rate is too low because $f_s/2 < B$.

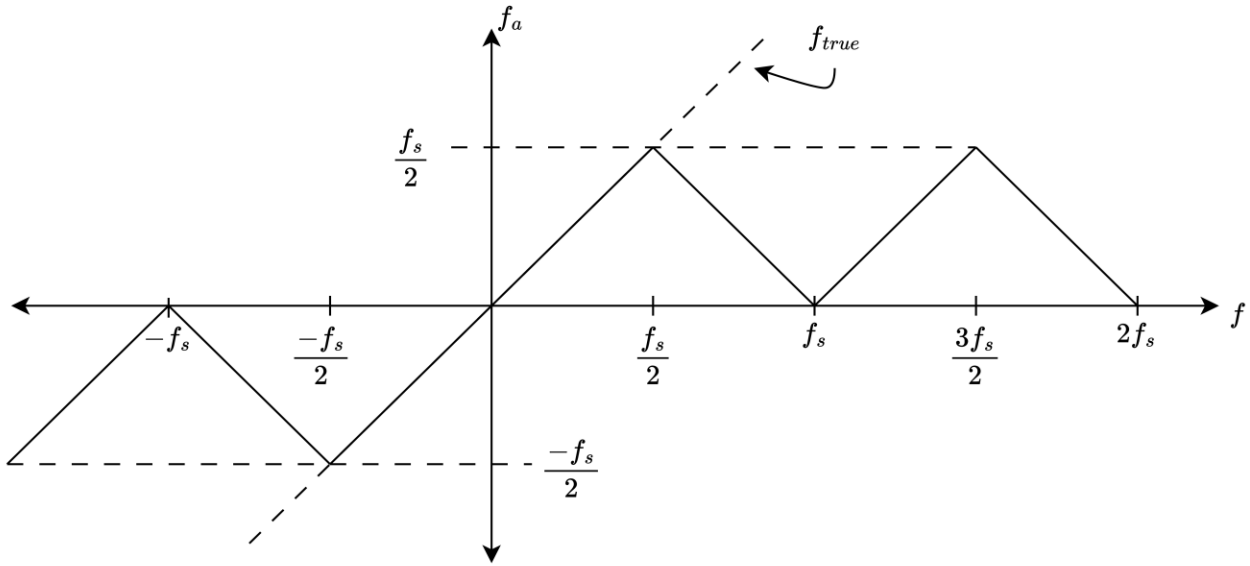
p.39 Richard G Lyons, Understanding Digital signal processing – 3rd edition

Aliasing

- Reconstructor – all values within the Nyquist interval $\left[{}^{-f_s}/_{2}, {}^{f_s}/_{2}\right]$ - LPF.
- The aliased frequency is given by

$$f_a = \begin{cases} f \bmod \left(\frac{f_s}{2}\right), & \text{if} \left\lfloor \frac{2f}{f_s} \right\rfloor = \text{even} \\ \left(\frac{f_s}{2}\right) - f \bmod \left(\frac{f_s}{2}\right), & \text{if} \left\lfloor \frac{2f}{f_s} \right\rfloor = \text{odd} \end{cases}$$





Hands-on

Baseband
and
Bandpass Signals

Definition: Baseband Signal

A baseband signal (m(t)) is a signal that occupies the frequency range from near 0 Hz (DC) up to a certain maximum frequency, without modulation.

- It is the original, unmodulated signal that carries information and is directly generated by a source.
- The term **baseband** is used to designate the frequency band of the original message signal from the source or the input transducer.

For Example:

- In telephony, the baseband is the audio band (band of voice signals) of 0 to 3.5 kHz.
- In FLAC, MP3 audio, the baseband is 0 Hz to 20 kHz.

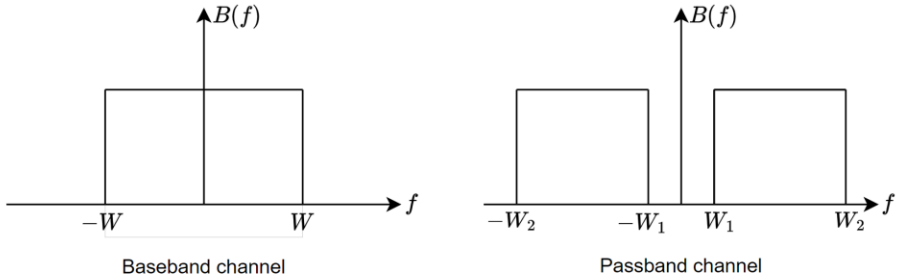
Baseband Signal(Cont.)

- In baseband communication, message signals are directly transmitted without any modification.
- Most baseband signals **contain significant low-frequency content**, they cannot be effectively transmitted over radio (wireless) links.
- Because **baseband signals have overlapping bands**, they would **interfere severely if sharing a common channel**. Thus, baseband communications leave much of the channel spectrum unused.
- By **modulating several baseband signals** and shifting their spectra to nonoverlapping bands, **many users can share one channel by utilizing most of the available bandwidth** through frequency division multiplexing **(FDM).**

Baseband Signal(Cont.)

- **Types of channel:**
  - Baseband channel – has frequency response of Low Pass Filter (LPF).
  - Passband channel – has frequency response of Band Pass Filter (BPF).

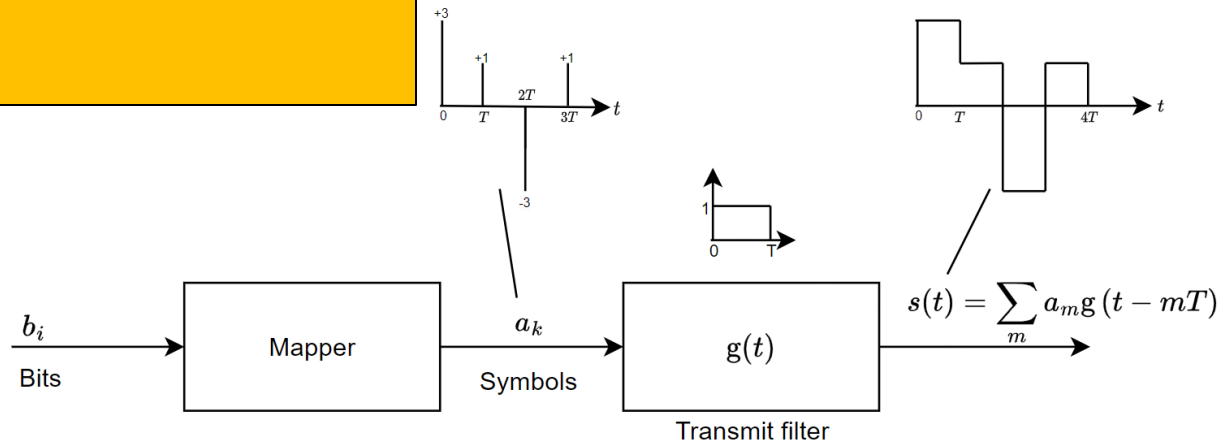
(a)
Channel



For Example:

- If we want to design baseband PAM signal,
  - The incoming bit are converted to modulating symbol stream by a mapper.
  - Restricted to finite alphabet $\mathcal{A}$. $a_k \in \mathcal{A}$ and $|\mathcal{A}| = 2^b$, for some integer b.
  - A mapper may map multiple bits into a single data symbol.

(b)
A PAM transmitter

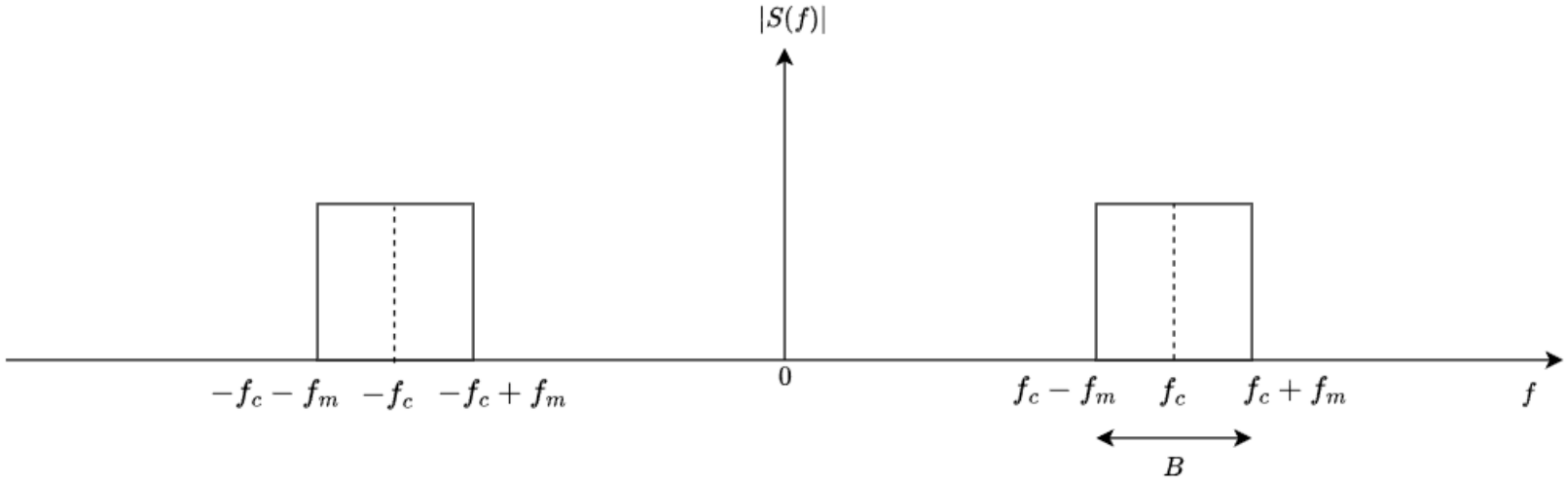


Definition: Bandpass Signal

A Bandpass Signal is a signal $s(t)$ whose Fourier transform $S(f)$ is nonzero only in some small band around some "central" frequency f_c.

For example:

$$S(f) = 0 \text{ for } |f - f_c| > f_m, \qquad \text{where} \qquad f_m < f_c$$



The **bandwidth** (B) of the bandpass signal = the width of the positive-frequency interval on which the signal is nonzero.

Bandpass Signal(Cont.)

- **I & Q form of baseband signal:**
  - Assuming that
    - $\tilde{s}(t)$ is complex –valued signal
    - Write its real and imaginary parts as

$$\tilde{s}(t) = I(t) + jQ(t)$$

where,

$I(t) \longrightarrow$ In-phase component

$Q(t) \longrightarrow$ Quadrature component

- **Relationship: I & Q parts and Bandpass signal:**

$$s_a(\mathrm{t}) = e^{j2\pi f_c t}\tilde{s}(t)$$

where,

- $s_a(\mathrm{t})$ is $\tilde{s}(t)$ shifted up

Bandpass Signal(Cont.)

- Using the I & Q form of $\tilde{s}(t)$

$$s_a(t) = e^{j2\pi f_c t}[I(t) + jQ(t)]$$

$$= [\cos(2\pi f_c \mathrm{t}) + j\sin(2\pi f_c \mathrm{t})][I(t) + jQ(t)]$$

$$= \boxed{[I(t)\cos(2\pi f_c \mathrm{t}) - Q(t)\sin(2\pi f_c \mathrm{t})]}$$

$$+j[I(t)\sin(2\pi f_c \mathrm{t}) + Q(t)\cos(2\pi f_c \mathrm{t})]$$

$$s_a(t) = \boxed{s(t)} + j\hat{s}(t)$$

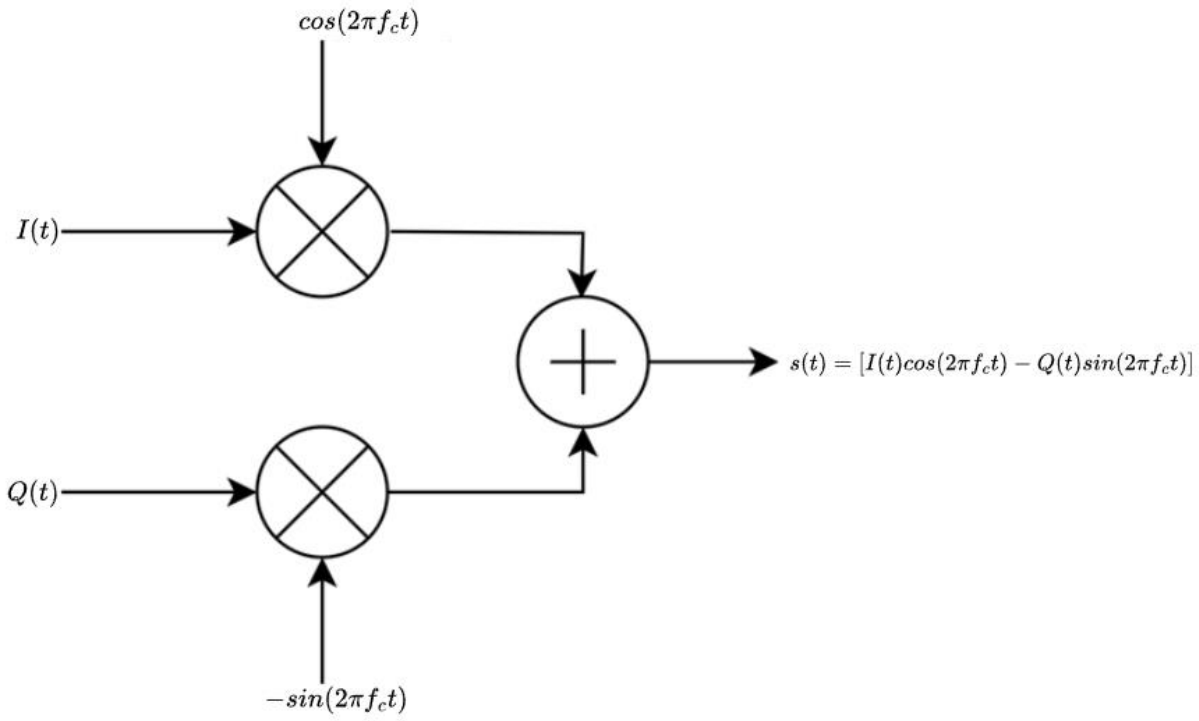
- This indicate how the I & Q components are related to BP signal.

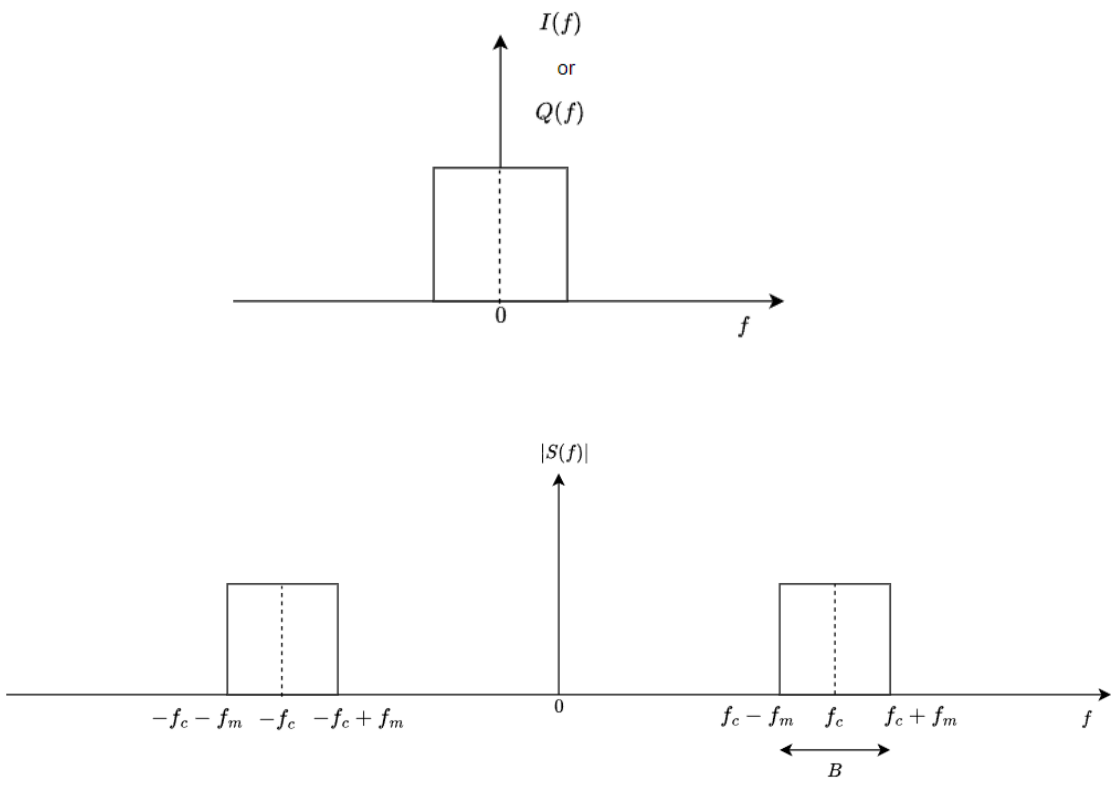
Bandpass Signal(Cont.)

- The in-phase component of $s_a(t)$ is denoted as

$$s(t) = Re\{\, s_a(\mathrm{t})\} = Re\{e^{j2\pi f_c t}\tilde{s}(t)\} = [I(t)\cos(2\pi f_c \mathrm{t}) - Q(t)\sin(2\pi f_c \mathrm{t})]$$

- **Design of Quadrature modulator:**





Magnitude and frequency plots of input and output waveforms of Quadrature modulator.

Bandpass Signal(Cont.)

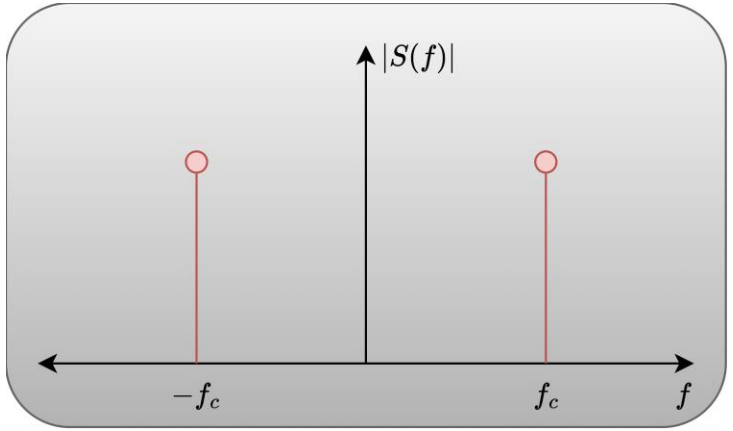
Analysis of $s(t)$ for different cases:

$$s(t) = [I(t)\cos(2\pi f_c \mathrm{t}) - Q(t)\sin(2\pi f_c \mathrm{t})]$$
$$s(t) = Re\{\tilde{s}(t)e^{j2\pi f_c t}\}$$

Case 1: $\tilde{s}(t) = acos(2\pi f_m t) + jbsin(2\pi f_m t) + c = 1 + 1j$, if $a = 0, b = 0$, and $c = 1 + 1j$, where $I(t) = 0, Q(t) = 0$.

$$s(t) = Re\{(1 + 1j)e^{j2\pi f_c t}\}$$
$$= Re\{(1 + 1j)[\cos(2\pi f_c \mathrm{t}) + j\sin(2\pi f_c \mathrm{t})]\}$$
$$= Re\{[\cos(2\pi f_c \mathrm{t}) - \sin(2\pi f_c \mathrm{t})] + j[\cos(2\pi f_c \mathrm{t}) + \sin(2\pi f_c \mathrm{t})]\}$$

$$s(t) = [\cos(2\pi f_c \mathrm{t}) - \sin(2\pi f_c \mathrm{t})]$$



Magnitude and Frequency plot of Quadrature modulator output.

Bandpass Signal(Cont.)

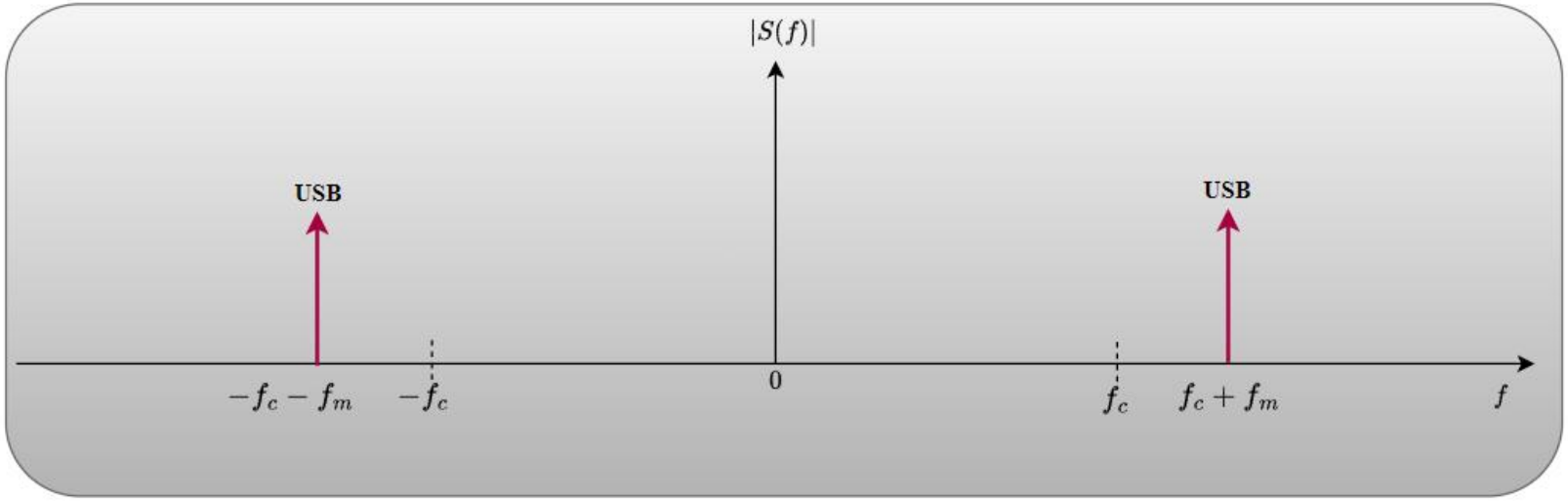
- **Case 2:** $\tilde{s}(t) = acos(2\pi f_m t) + jbsin(2\pi f_m t) + c = cos(2\pi f_m t) + jsin(2\pi f_m t)$, if $a = 1, b = 1, c = 0$, where $I(t) = cos(2\pi f_m t), Q(t) = sin(2\pi f_m t)$.

$$s(t) = Re\{[cos(2\pi f_m t) + jsin(2\pi f_m t)]e^{j2\pi f_c t}\}$$

$$= Re\{[cos(2\pi f_m t) + jsin(2\pi f_m t)][\cos(2\pi f_c t) + j\sin(2\pi f_c t)]\}$$

$$s(t) = [cos(2\pi f_m t)\cos(2\pi f_c t) - sin(2\pi f_m t)\sin(2\pi f_c t)] = \boxed{\cos(2\pi(f_c + f_m)t)}$$

Upper side band (USB)



Magnitude and Frequency plot of Quadrature modulator output.

Bandpass Signal(Cont.)

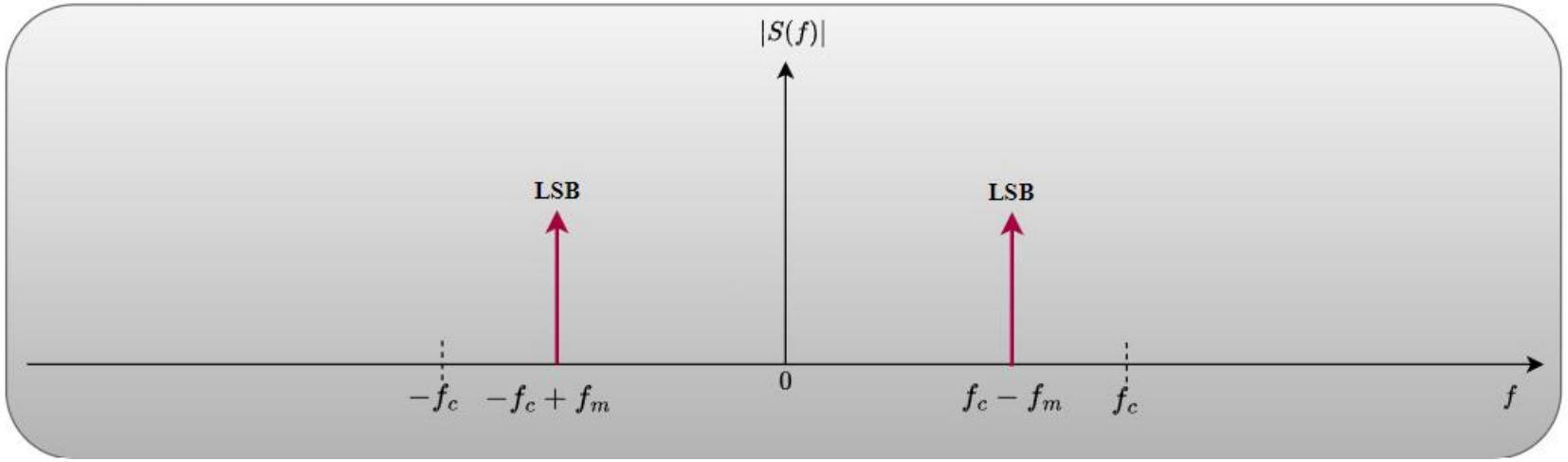
- **Case 3:** $\tilde{s}(t) = acos(2\pi f_m t) + jbsin(2\pi f_m t) + c = cos(2\pi f_m t) - jsin(2\pi f_m t)$, if $a = 1, b = -1, c = 0$, where $I(t) = cos(2\pi f_m t), Q(t) = -sin(2\pi f_m t)$.

$$s(t) = Re\{[cos(2\pi f_m t) - jsin(2\pi f_m t)]e^{j2\pi f_c t}\}$$

$$= Re\{[cos(2\pi f_m t) - jsin(2\pi f_m t)][\cos(2\pi f_c t) + j\sin(2\pi f_c t)]\}$$

$$s(t) = [cos(2\pi f_m t)\cos(2\pi f_c t) + sin(2\pi f_m t)\sin(2\pi f_c t)] = \boxed{\cos(2\pi(f_c - f_m)t)}$$

Lower side band (USB)



Magnitude and Frequency plot of Quadrature modulator output.

Bandpass Signal(Cont.)

- **Case 4:** $\tilde{s}(t) = acos(2\pi f_m t) + jbsin(2\pi f_m t) + c = cos(2\pi f_m t) + 1$, if $a = 1, b = 0$, $c = 1$, where $I(t) = cos(2\pi f_m t), Q(t) = 0$.

$$s(t) = Re\{[1 + cos(2\pi f_m t)]e^{j2\pi f_c t}\}$$

$$= Re\{[1 + cos(2\pi f_m t)][\cos(2\pi f_c t) + j\sin(2\pi f_c t)]\}$$

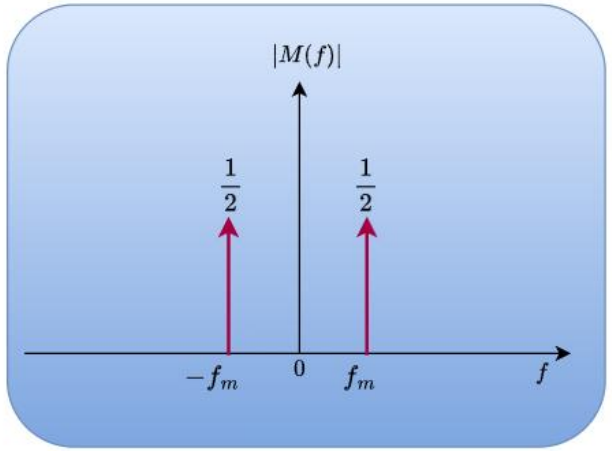
$$= Re\{[1 + cos(2\pi f_m t)]\cos(2\pi f_c t) + j[1 + cos(2\pi f_m t)]\sin(2\pi f_c t)\}$$

$$s(t) = [1 + cos(2\pi f_m t)]\cos(2\pi f_c t) \longrightarrow$$ **Amplitude Modulation (AM)**

- **Assume that**

$$m(t) = cos(2\pi f_m t)$$

$$m(t) \xrightarrow[\text{F.T.}]{} M(f) = \frac{1}{2}[\delta(f - f_m) + \delta(f + f_m)]$$



Frequency spectrum

Bandpass Signal(Cont.)

$$s(t) = [1 + cos(2\pi f_m t)]\cos(2\pi f_c t)$$

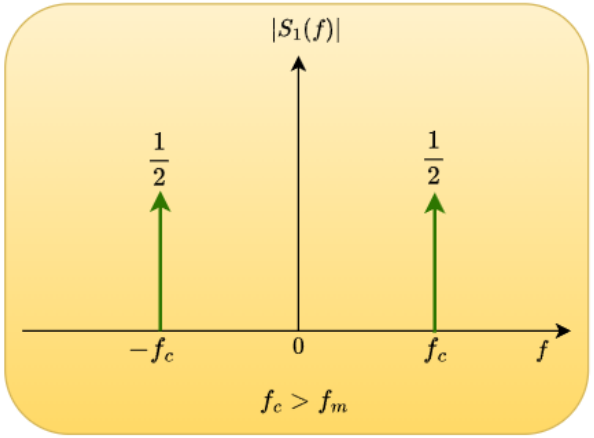
Amplitude Modulation (AM)

$$s(t) = [\cos(2\pi f_c t) + cos(2\pi f_m t)\cos(2\pi f_c t)]$$

F.T.

$$S(f) = \left[\frac{1}{2}\left(\delta(f - f_c) + \delta(f + f_c)\right)\right] \longrightarrow s_1(t)$$

$$+\frac{1}{4}\left(\delta(f - (f_c+f_m) + \delta(f + (f_c+f_m) + \delta(f - (f_c-f_m) + \delta(f + (f_c-f_m))\right)\Big] \longrightarrow s_2(t)$$



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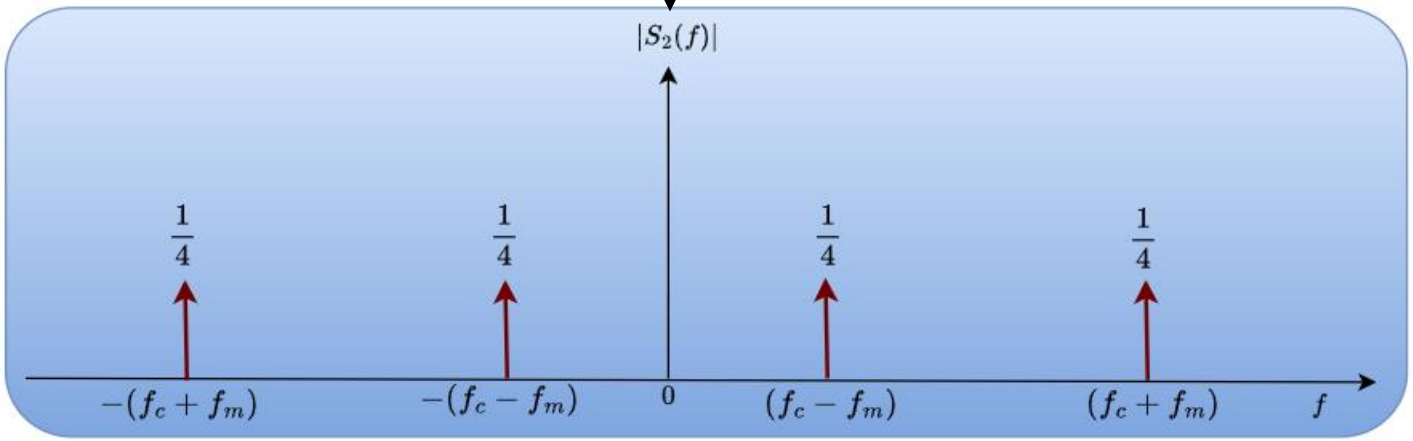
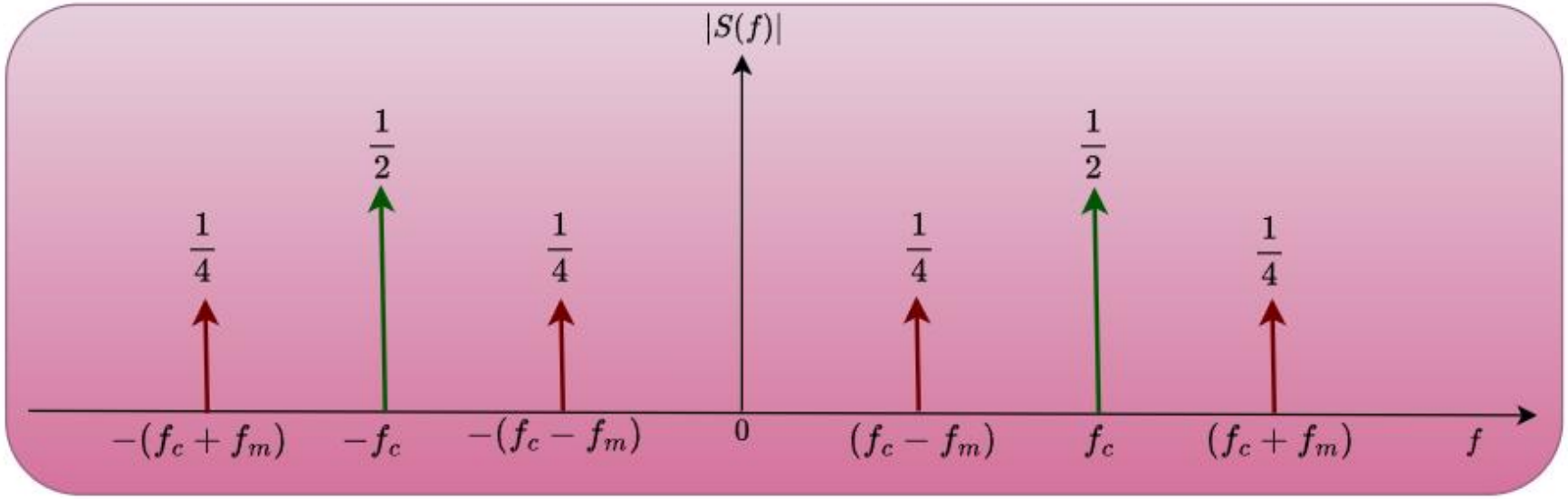


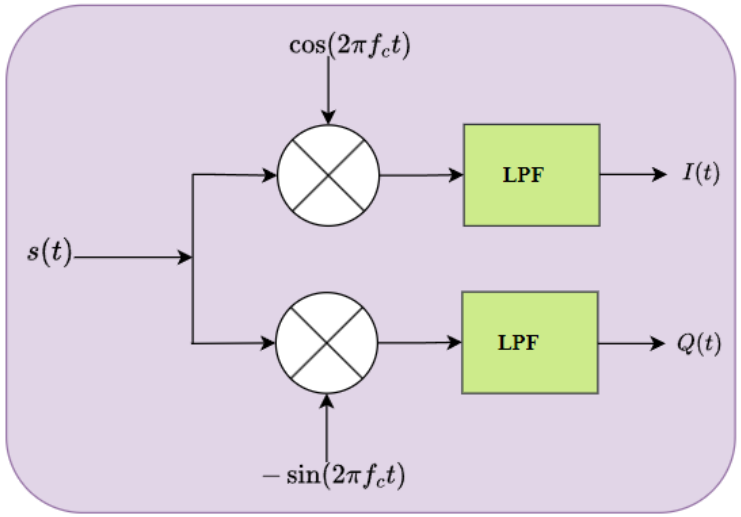
Bandpass Signal(Cont.)

$$S(f) = \left[\frac{1}{2}\left(\delta(f - f_c) + \delta(f + f_c)\right) + \frac{1}{4}\left(\delta(f - (f_c + f_m) + \delta(f + (f_c + f_m) + \delta(f - (f_c - f_m) + \delta(f + (f_c - f_m)\right)\right]$$

Frequency spectrum of $S(f)$:



Bandpass Signal(Cont.)



Quadrature demodulator

- **Design of Quadrature demodulator:**

$$\boxed{\textbf{If } s(t) = [1 + cos(2\pi f_m t)]cos(2\pi f_c t)}$$

$$I(t) = s(t)\cos(2\pi f_c t) = ([1 + cos(2\pi f_m t)]\cos(2\pi f_c t))\cos(2\pi f_c t)$$

$$= [1 + cos(2\pi f_m t)]cos^2(2\pi f_c t) = [1 + cos(2\pi f_m t)]\left(\frac{1}{2}[1 + \cos(2\pi f_c t)\right)$$

$$= \frac{1}{2} + \frac{1}{2}[cos(2\pi f_m t)] + \frac{1}{2}[cos(4\pi f_c t)] + \frac{1}{4}[cos(2\pi(2f_c + f_m)t)] + \frac{1}{4}[cos(2\pi(2f_c - f_m)t)]$$

$$\boxed{\textbf{LPF } (f \leq f_c)}$$

$$\boxed{\frac{1}{2} + \frac{1}{2}[cos(2\pi f_m t)]}$$

- **Passing through the low pass filter (LPF):**

A low-pass filter allows only low-frequency components (e.g. $f \leq f_c$) to pass through

- The DC component $\frac{1}{2}$ will pass.
- The term $\frac{1}{2}[cos(2\pi f_m t)]$ will pass if f_m is within the filter cutoff.
- High-frequency components $2f_c$ and $2f_c \pm f_m$ will be attenuated.

Hands on